Risk Minimization Framework.

Given Data $D = \left\{ (x_i, y_i) \right\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{xy}$

$$x \in \mathbb{R}^d \quad y \in \mathbb{R}^k / \{1, 2, 3, \ldots k\}$$

Goal is to estimate a hypothesis fn.

$h : \mathcal{X} \longrightarrow \mathcal{Y}$, $h \in \mathcal{H}$: set of all functions from $\mathcal{X} \to \mathcal{Y}$.

Quantify the "goodness" of an $h(x)$.

Define a notion of Loss-function L.

$$L : \mathcal{Y} \times \mathcal{Y} \longrightarrow \mathbb{R}^+$$

$$L\left(h(x), y\right) \longrightarrow \mathbb{R}^+$$

Eg: a) $L\left(h(x), y\right) = \|h(x) - y\|_2^2$: sq. error loss

b) $L\left(h(x), y\right) = \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{if } h(x) \neq y \end{cases}$: zero-one loss.

Extending the loss fn for the population.

Define $R(h) \triangleq \mathbb{E}_{\mathbb{P}_{xy}} L\big(h(x), y\big)$: Risk function.

problem : Estimate $h^*(x)$

$$h^*(x) = \underset{h \in \mathcal{H}}{\operatorname{argmin}} \left[\mathbb{E}_{\mathbb{P}_{xy}} L\big(h(x), y\big) \right]$$

Risk-minimization framework.

True risk function can't be minimized

$\because$ $\mathbb{P}_{xy}$ is unknown.

$\therefore$ Consider a surrogate

$$\hat{R}(h) \triangleq \frac{1}{N} \sum_{i=1}^{N} L\big(h(x_i), y_i\big)$$: Empirical Risk.

$$\boxed{\hat{h}^*(x) = \underset{h \in \mathcal{H}}{\operatorname{argmin}}\ \hat{R}(h)}$$

$x_i, y_i \overset{iid}{\sim} \mathbb{P}_{xy}$

Empirical Risk Minimization (ERM)

We know, $\hat{R}(h) \longrightarrow R(h)$: LLNs.

However, $\hat{h}^*(x)$ may not converge to $h^*(x)$.

Central question : When would $\hat{h}^*(x) \to h^*(x)$?

Generalization.

<u>Example of ERM for a Regression problem.</u>

Given $D = \{(x_i, y_i)\}_{i=1}^N \sim \text{iid } P_{xy}$

$x_i \in \mathbb{R}^d, \quad y \in \mathbb{R}$

Suppose $h_\theta(x)$ comes from a family of parametric functions, parameterized by θ.

$h_\theta(x) = \theta^T x \qquad h_\theta(x) = e^{\theta^T x}$

Define Loss fn, mean sq. error

$$\hat{R}(h_\theta) = \frac{1}{N} \sum_{i=1}^N \left[h_\theta(x_i) - y_i \right]^2$$

$$\theta^* = \underset{\theta}{\arg\min}\ \hat{R}(h_\theta)$$

The above is equivalent to estimating $p_\theta(y|x)$ using ML estimation with $p_\theta(y|x) \sim \mathcal{N}(y ; h_\theta(x), I)$

Equivalence between Risk minimization

&

Divergence Minimization (Max. likelihood)

<u>Question</u>: What function minimizes the true risk?

Consider $D = \{(x_i, y_i)\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{xy}$

$x_i \in \mathbb{R}^d \qquad y_i \in \{0, 1\}$

$$L(h(x), y) = \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{otherwise} \end{cases}$$

$$R(h) = \mathbb{E}_{\mathbb{P}_{xy}} L(h(x), y)$$

Question : What h, minimizes the true risk for the above scenario?

Ans: Consider a hypothesis fn. as follows:

$$h_B(x) = \begin{cases} 0 & \text{if } P(y=0|x) > P(y=1|x) \\ 1 & \text{if } P(y=1|x) \geq P(y=0|x) \end{cases}$$

→ Baye's Classifier.

Claim: $R(h_B) \leq R(h)$

$\forall h \in \mathcal{H}$.

proof: Let $S_i(h) = \{x \in \mathbb{R}^d : h(x) = i\}$

$i = 0, 1$.

$$S_0(h) \cap S_1(h) = \phi$$

$$S_0(h) \cup S_1(h) = \mathbb{R}^d \quad \forall h \in \mathcal{H}.$$

Note $R(h) = \mathbb{P}\left(I_{h(x) \neq y}\right)$

where $I_{h(x) \neq y}$ is an indicator Bernoullic RV.

$$R(h) = \mathbb{E}_{\mathbb{P}_{xy}}\left[L\left(h(x), y\right) \right]$$

$$= \mathbb{P}\left(\mathbb{I}_{h(x) \neq y} \right)$$

$$= \mathbb{P}\left(h(x) = 1, \; y = 0 \right) + \mathbb{P}\left(h(x) = 0, y = 1 \right)$$

$$= \mathbb{P}\left(x \in S_1(h), \; y = 0 \right) + \mathbb{P}\left(x \in S_0(h), y = 1 \right)$$

$$= \mathbb{P}(y=0) \cdot \mathbb{P}_{x|y}\left(x \in S_1(h) \,\middle|\, y = 0 \right) +$$

$$\mathbb{P}(y=1) \; \mathbb{P}_{x|y}\left(x \in S_0(h) \,\middle|\, y = 1 \right)$$

$$= \mathbb{P}(y=0) \int_{S_1(h)} p_{x|y=0}\left(x \middle| y = 0 \right) dx + \mathbb{P}(y=1) \int_{S_0(h)} p_{x|y=1} \, dx$$

since $S_1(h) \cap S_0(h) = \phi$,

$\forall \, x \in \mathbb{R}^d$, we only consider one of the two integrals.

for the case of $h_B(x)$,

$$R(h) = \int_{S_1} \mathbb{P}_{y=0} \cdot P_{x|y=0} \, dx + \int_{S_0} \mathbb{P}_{y=1} \cdot P_{x|y=1} \, dx$$

$$h_B = \begin{cases} 1 & \text{if } P_{y=1|x} > P_{y=0|x} \Rightarrow P_{x|y=1} \cdot P_{y=1} > P_{x|y=0} \cdot P_{y=0} \\ 0 & \text{if } P_{x|y=0} \cdot P_{y=0} \geq P_{x|y=1} \cdot P_{y=1} \end{cases}$$

$$\Rightarrow R(h_B) = \int_{R^d} \min \left(P_{y=0} \cdot P_{x|y=0} , P_{y=1} \cdot P_{x|y=1} \right) dx$$

$$\therefore \boxed{R(h_B) < R(h) \quad \forall h \in \mathcal{H}}$$

$$\underline{\mathbb{E}_{y|x}} = \underset{h}{\arg\min} \; \mathbb{E}_{P_{xy}} \left[\left(h(x) - y \right)^2 \right]$$